NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

What Does This Test Support?

AP Statistics · Unit 4 · Topic 4.10 · B: 2027-01-29 · E: 2027-03-19

[STAR] TODAY'S PROBLEM

Suppose this term 4,000 students chose Focus mode and 3,600 chose Guided. Independent random samples of 100 students from each mode completed the same practice set. The service asked in advance whether mean Focus time was greater and set $\alpha = 0.05$. Only a difference of at least 3 minutes matters in practice.

Nia: "The $p \approx 0.0263 < 0.05$ supports a higher Focus mean. So Guided causes every student to finish at least 3 minutes faster; switch everyone."

Mateo: "The sample gap is 1.2 minutes, and students chose their modes. A 3-minute effect and causation are not established. So retain H_0; there is no evidence of different population means."

Practice time summaries (minutes)

Mode	N	n	$\bar{x}$	s
Focus	4,000	100	42.8	4.2
Guided	3,600	100	41.6	4.5

FIRST TAKE — before the video (5 min)

Which claim seems too strong: no average difference, or Guided helps everyone? Give one reason.

[BOOK] CARRY OUT THE TEST, THEN BOUND THE CONCLUSION (keep on the page)

For independent samples, $SE = \sqrt{s_1^2/n_1 + s_2^2/n_2}$ and $t = ((\bar{x}_1 - \bar{x}_2) - 0)/SE$. A p-value is the probability, assuming H_0, of a statistic at least as extreme in the direction of H_a. If $p \leq \alpha$, reject H_0 and state evidence for H_a in context. Significance against 0 does not establish practical size or an individual guarantee. Random sampling supports population scope; random assignment, not self-selection, supports causation.

Finish after the video:

DOK 1 **(a)** State H_0 and H_a for the population mean times in Focus-minus-Guided order.

DOK 2 **(b)** The difference is 1.2 minutes and $SE = 0.6155485$. Calculate t and interpret the supplied one-sided $p = 0.0263$ in one sentence.

DOK 3 **(c)** In 4 sentences, combine the supported parts of both reports. Use the p-value for the mean claim, compare 1.2 with the 3-minute benchmark, and explain why self-selection and a mean difference do not justify switching every student.

Frame: The test supports ____________________ because ____________________. The 3-minute claim is ____________ because ____________________.

Frame: We cannot promise this for every student because ____________________; mode choice also limits ____________________.

EXIT — turn this in

One thing the video changed about my first take: ____